GS FOUNDATION 2024-25

GEOGRAPHY - 2

Motions of the Earth

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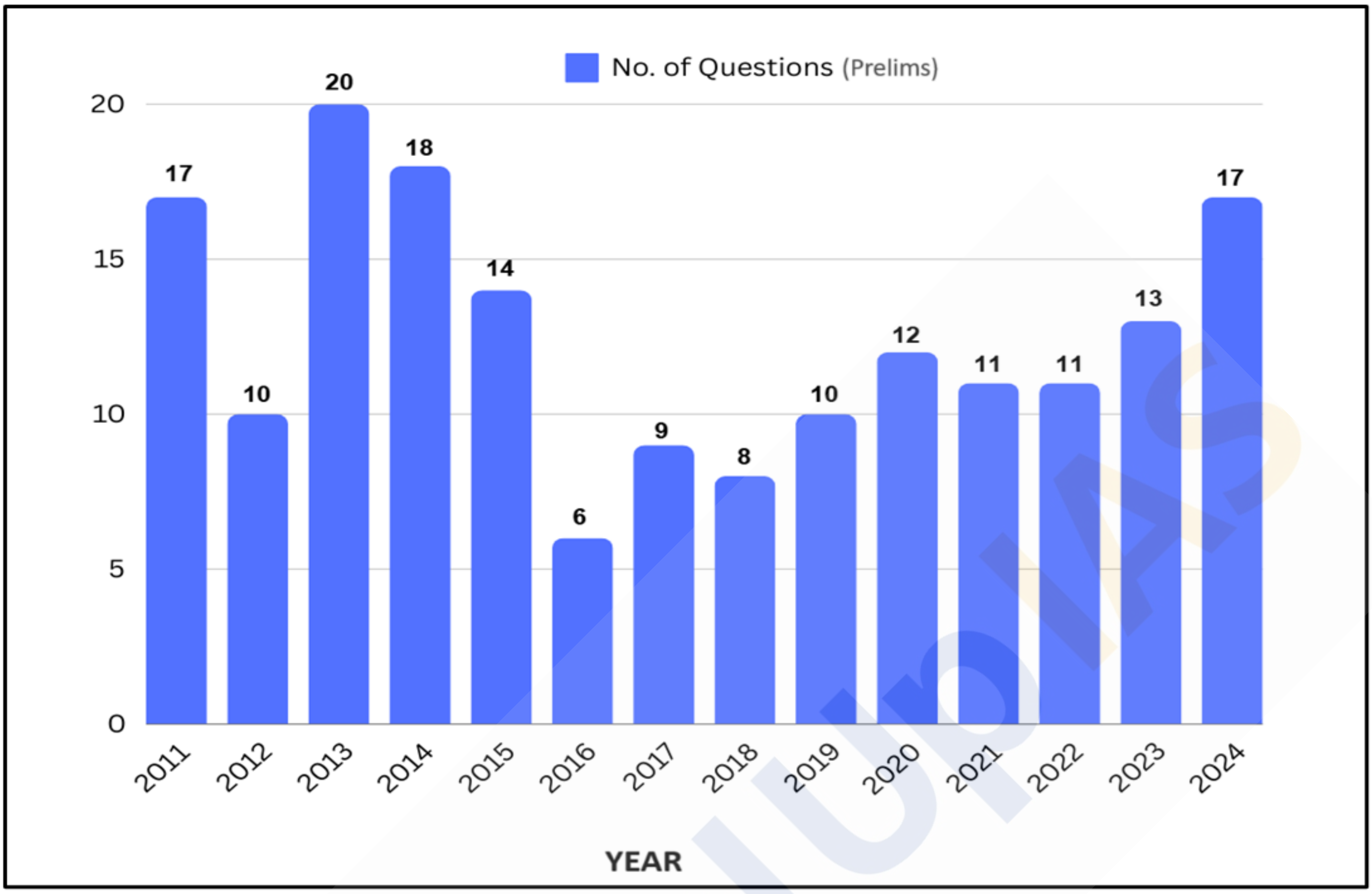

Topic-wise Trend Analysis (2011 – 2024)

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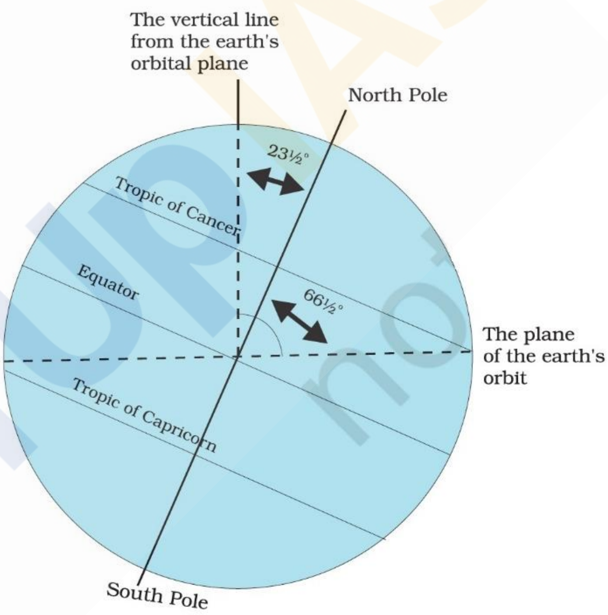
Motions of the Earth

Introduction:

The Earth, our dynamic planet, is constantly in motion. **Two fundamental movements define its journey through space - Rotation and Revolution.** Rotation refers to the Earth's spinning on its axis, while revolution describes its orbit around the Sun. These motions are not merely astronomical phenomena but have profound effects on our daily lives, influencing the cycle of day and night, the progression of seasons, and the global climate patterns. Understanding these motions provides crucial insights into the functioning of the Earth as a system and its interactions with the broader cosmos.

Rotation of the Earth:

The Earth rotates from **west to east** on an **imaginary axis** that runs from the North Pole to the South Pole. This rotation takes approximately **24 hours** to complete, resulting in the **cycle of day and night.**



Earth rotates on a **tilted axis.** Earth's rotational axis makes an angle of **23.5° with the normal** i.e. it makes an angle of **66.5° with the orbital plane.** Orbital plane is the plane of earth's orbit around the Sun. This tilt is responsible for the variation in the length of daylight and the changing position of the Sun in the sky throughout the year.

Key aspects of Earth's rotation include:

- **Speed of Rotation -** The Earth rotates at a speed of about **1670 kilometers per hour** (1037 miles per hour) **at the equator.** The speed **decreases as one moves towards the poles.**
- **Coriolis Effect -** Due to the rotation of the Earth, moving air and water are **deflected to the right in the Northern Hemisphere** and to the **left in the Southern Hemisphere.** This effect is crucial in shaping weather patterns and ocean currents.
- **Diurnal Cycle -** The rotation causes the diurnal cycle, the pattern of **daylight and darkness** that influences biological rhythms, human activities, and ecosystems.

Do you know?

If the **Earth did not rotate,** the side of the Earth facing the Sun would experience constant **daylight for approximately six months,** thus would lead to extreme heating and potentially inhospitable temperatures. The side facing away from the Sun would be in **darkness for about six months,** thus would result in extreme cooling, possibly leading to freezing conditions. **Life would not have been possible in such extreme conditions.**

Revolution:

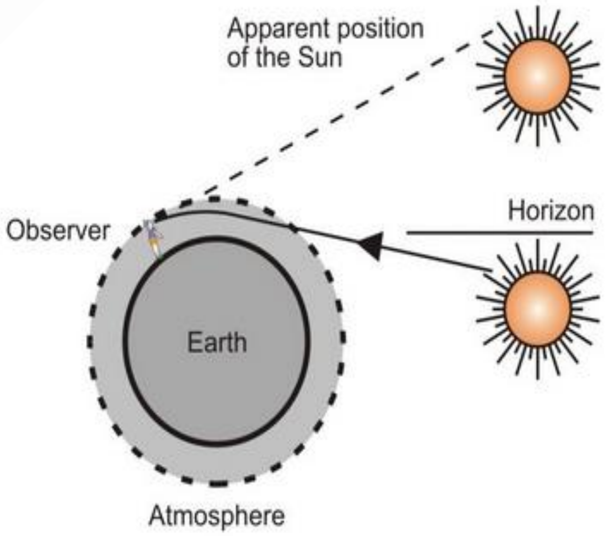
The Earth's revolution is its **movement around the Sun in an elliptical orbit,** taking approximately **365.25 days** to complete. This orbital journey defines the length of a year and is responsible for the progression of the seasons.

Key aspects of Earth's revolution include:

- **Orbital Path -** The Earth's orbit is an **ellipse,** with the Sun located at one of the two foci. This elliptical path means that the Earth is not always at the same distance from the Sun. The **closest point is called perihelion,** and the **farthest point is aphelion.**
- **Tilt and Seasons -** The axial tilt of the Earth remains pointed in the same direction as it orbits the Sun, leading to **varying angles of sunlight at different times of the year.** This variation causes the seasons—**spring, summer, autumn, and winter.** During summer in the Northern Hemisphere, the North Pole is tilted towards the Sun, resulting in longer days and shorter nights.
- **Solstices and Equinoxes -** The **solstices** mark the points in the Earth's orbit where the tilt of the axis is most pronounced towards or away from the Sun, leading to the **longest and shortest days of the year.** Solstices are the points in the Earth's orbit where the **Sun is at its greatest distance from the equatorial plane.**
  - **Equinoxes** are the points in the Earth's orbit where the **Sun is directly above the equator,** resulting in nearly **equal day and night durations globally.**

Why are days always longer than nights at the equator?

Days are always longer than nights at the equator primarily due to **atmospheric refraction.** If there were no atmosphere, the lengths of day and night would be nearly equal at the equator, especially during equinoxes. However, the **atmosphere bends the sun's rays, particularly during sunrise and sunset when the rays are slant, causing the apparent image of the sun to appear above the horizon even when it is below it.**



Atmospheric refraction effects at sunrise and sunset

Additionally, **at the equator,** the length of day and night remains nearly equal throughout the year due to the **minimal effect of the Earth's axial tilt and the Sun's vertical path across the sky.** During equinoxes, the Sun is directly overhead at the equator, making day and night approximately 12 hours each. However, the bending of light due to atmospheric refraction extends daylight, making days slightly longer than nights.

Why does the temperature decrease as we move from the equator toward the poles?

First, the angle at which the Sun's rays strike the Earth changes; **at higher latitudes, sunlight hits the Earth at a lower angle, spreading the solar energy over a larger area,** which diminishes its intensity.

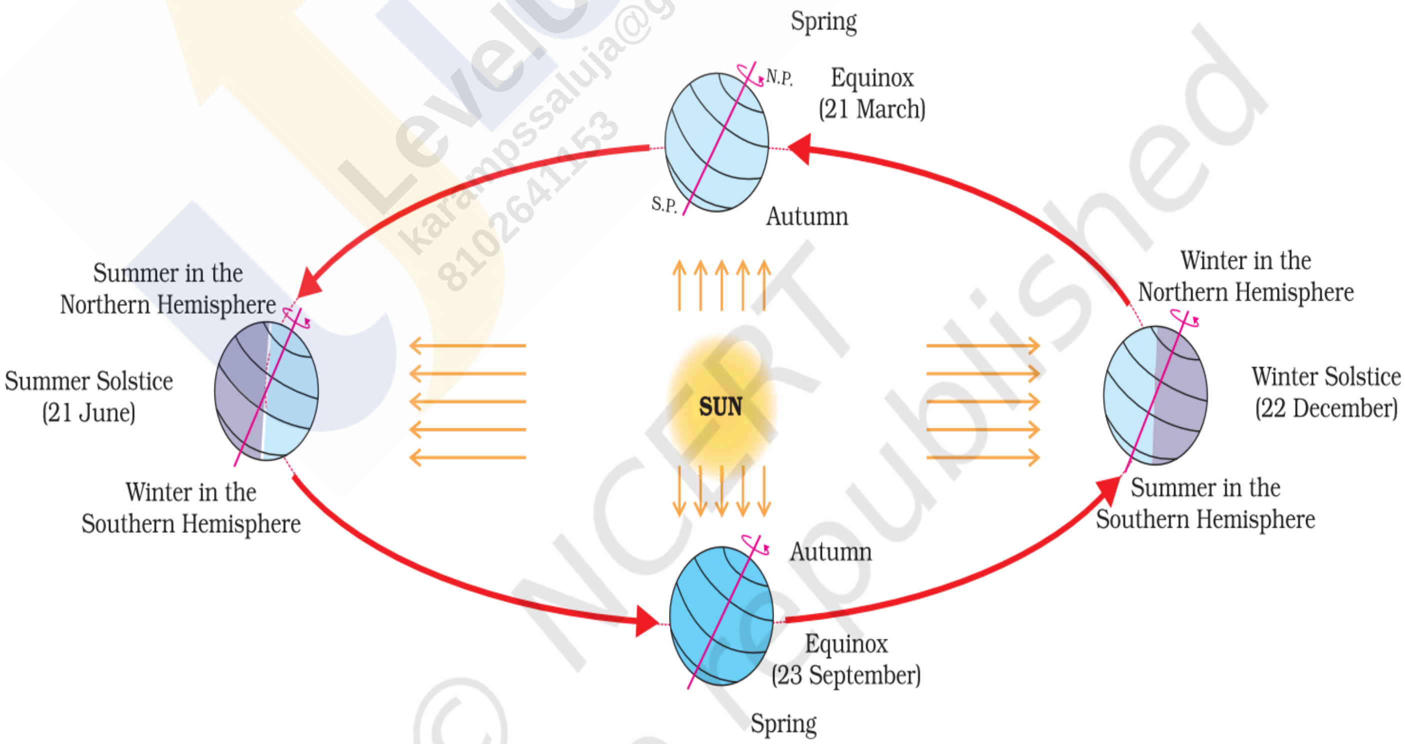
Additionally, sunlight traveling through a greater atmospheric path length at higher latitudes results in **more absorption and scattering of solar radiation, reducing the amount of energy that reaches the surface.**

The **albedo effect** also plays a significant role as the **polar regions, covered with snow and ice, reflect a substantial portion of solar radiation back into space** due to their high reflectivity, keeping the surface cooler.

Solstice:

Solstices are the points in the Earth's orbit where the **Sun is at its greatest distance from the equatorial plane.** A solstice occurs **twice a year,** marking the days with the **shortest and longest durations of sunlight in a hemisphere. There are two types of solstices:**

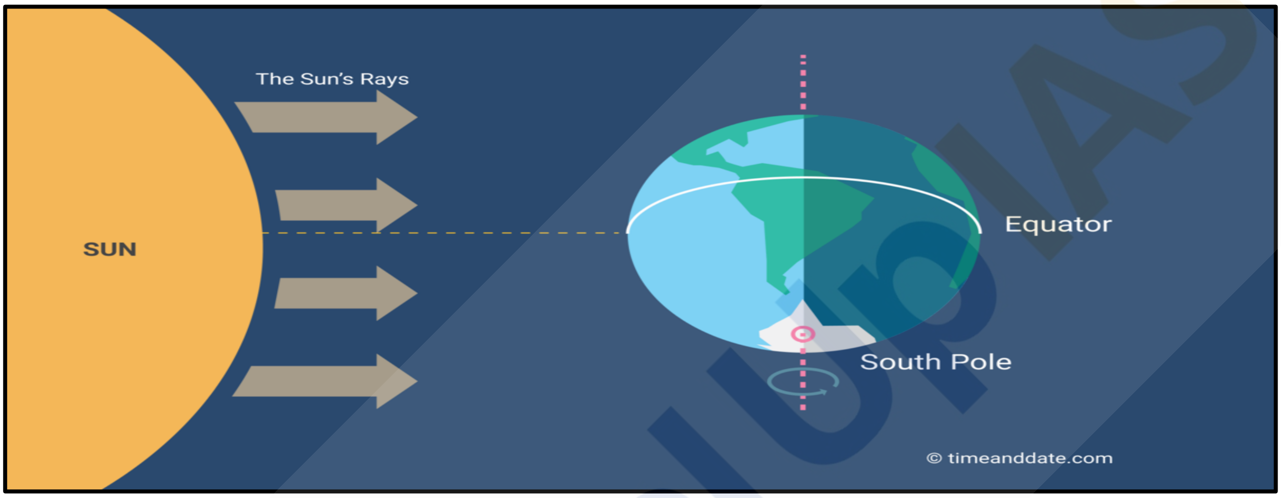
- **Summer Solstice:** Occurs around **June 21** when the **Sun is directly overhead at the Tropic of Cancer.** It occurs when **North Pole is tilted closest to the Sun.** It marks the **longest day and shortest night** of the year in the **Northern Hemisphere.** The places beyond the Arctic circle experience continuous daylight for about six months. As a large portion of Northern Hemisphere receives sunlight and heat during summer solstice, it is summers in Northern Hemisphere; whereas winters in Southern Hemisphere. Although the summer solstice is the longest day of the year for the Northern Hemisphere, the dates of the earliest sunrise and latest sunset vary by a few days. This is because the Earth orbits the Sun in an ellipse, and its orbital speed varies slightly during the year
- **Winter Solstice:** Occurs around **December 21** when the **Sun is directly overhead at the Tropic of Capricorn.** It occurs when **North Pole is tilted farthest from the Sun.** It marks the **shortest day and longest night** of the year in the **Northern Hemisphere.** The places beyond the Antarctic circle experience continuous daylight for about six months. As a large portion of Southern Hemisphere receives sunlight and heat during winter solstice, it is summer in Southern Hemisphere, whereas winter in Northern Hemisphere. Although the winter solstice itself lasts only a moment, the term sometimes refers to the day on which it occurs.



Equinox:

The word equinox is derived from two Latin words – **aequus** (equal) and **nox** (night). Equinox refers to a day with an **equal duration of day and night.** There are **only two times of the year** when the **Earth's axis is tilted neither toward nor away from the sun,** resulting in a nearly equal amount of daylight and darkness at all latitudes. These events are referred to as **Equinoxes.**

The equinoxes happen in **March (about March 21)** and **September (about September 23).** These are the days when the **Sun is exactly above the Equator,** which makes day and night of equal length. It can be noted that the most places on Earth receive more than 12 hours of daylight on equinoxes. This is because of the atmospheric refraction of sunlight and how the length of the day is defined. The equinoxes are prime time for Northern Lights – geomagnetic activities are twice more likely to take place in the spring and fall time, than in the summer or winter.



Why regions beyond the Arctic Circle receive sunlight all day long in summer?

Regions beyond the Arctic Circle experience continuous daylight during the summer months due to the **tilt of the Earth's axis.** In this period, the **North Pole is tilted toward the Sun,** allowing these regions to receive sunlight all day long. This phenomenon, known as the **Midnight Sun,** occurs from late May to late July, peaking around the Summer Solstice when the **Sun does not set and remains visible for 24 hours.** The duration of the Midnight Sun increases with latitude, reaching up to six months at the North Pole.